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(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# Wireless Communication

## ITA0321– MOBILE COMPUTING

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# Introduction & Problem Statement

## Introduction

1. Wireless communication transfers data between devices without physical cables using electromagnetic waves.
2. It is used in mobile phones, Wi-Fi, Bluetooth, and satellite communication.

## Problem Statement

1. Wireless communication faces challenges such as signal interference, security risks, and limited bandwidth.
2. A reliable wireless system is needed to provide secure, fast, and uninterrupted connectivity

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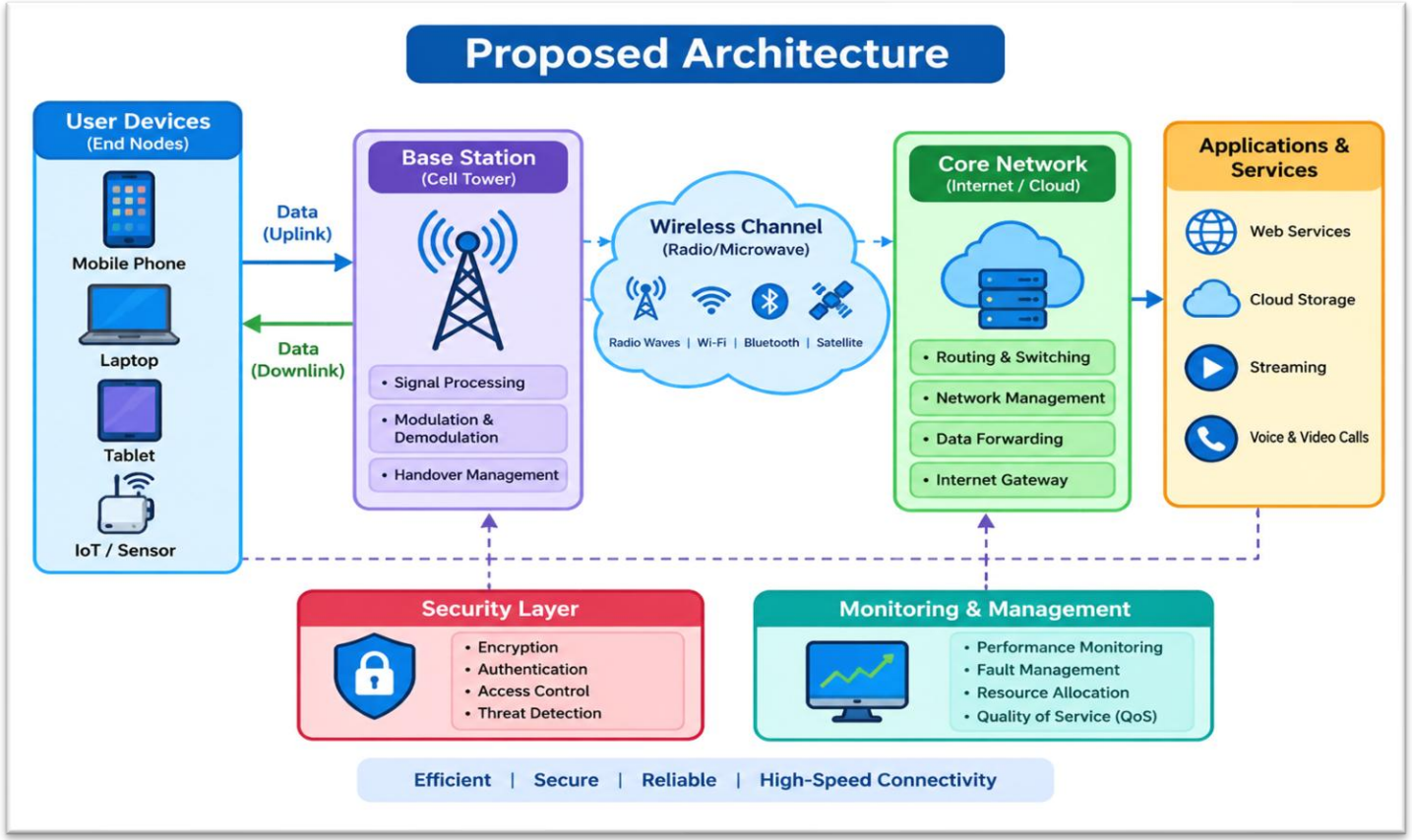
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# Proposed Architecture



# Wireless Transmission & Propagation

**Wireless Transmission:** The transfer of data between devices without physical cables using electromagnetic waves.

**Example:** Sending data through Wi-Fi, Bluetooth, or mobile networks without cables.

**Wireless Propagation**: The movement of radio waves through air or space from a transmitter to a receiver via reflection, diffraction, and scattering.

**Example:** Radio signals traveling from a cell tower to a mobile phone.

# Conclusion & Future Scope

## Conclusion

- Wireless communication enables fast and reliable data transmission without physical cables.
- It supports mobile networks, Wi-Fi, IoT devices, and various digital services.

## Future Scope

- 6G technology will improve data speeds, connectivity, and network performance.
- AI, IoT, and satellite communication will enable smarter and wider wireless connectivity.